INCH-POUND
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MILITARY SPECIFICATION

MOLDING MATERIAL, GLASS, EPOXY IMPREGNATED

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of epoxy impregnated glass molding material.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN; AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-52770 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330
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SPECIFICATIONS

MILITARY

MIL-R-9300	-	Resin, Epoxy, Low Pressure Laminating
MIL-R-60346	-	Roving, Glass, Fibrous (for Prepreg Tape and Rotating, Filament Winding, and Pultrusion Applications)

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF TESTING AND MATERIALS

ASTM C 613	-	Resin Inorganic-Filled Plastics
ASTM C 695	-	Compressive Properties of Rigid Plastics
ASTM D 695	-	Resin in Inorganic-Filled Plastics
ASTM D 792	-	Specific Gravity for Displacement of Water
ASTM D 3951	-	Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified (see 6.2), a first article sample shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Composition. The material shall be a pre-mix consisting of a glass reinforcement impregnated with an epoxy resin system (see 4.5.4.1).

3.2.1.1 Reinforcement. The reinforcement shall meet the requirements of MIL-R-60346, Type I, K-37 Glass, Class 1, except that it shall be chopped. Reinforcement shall consist of 50 ± 5% by weight of 0.250 ± 0.062 inch long fibers and 50 ± 5% by weight of 0.125 ± 0.062 inch long fibers (see 4.5.4).

3.2.1.2 Resin system. The resin system shall consist of epoxy resin conforming to MIL-R-9300, Type I, Grade O, Form A, modified with a suitable elastomer (see 4.5.4.4).

3.2.2 Storage life. The material as furnished by the manufacturer shall be capable of meeting the requirements specified herein after storage at a maximum temperature of + 4.4º Celsius (C) for not less than 90 days (see 4.5.4.2).

3.3 Performance characteristics.

3.3.1 Physical and chemical properties. The physical and chemical properties of the material shall be as specified in Table I.

TABLE I. Physical and chemical properties.

Characteristics	Average		Individual	
	Minimum	Maximum	Minimum	Maximum
Resin content, % by weight	42	52	40	55
Volatile content, % by weight	--	0.5	--	0.8
Specific gravity	1.50	--	1.48	--
Gel time, minutes	2.5	5.0	2.0	5.5

3.3.2 Mechanical properties. The mechanical properties of the material shall be as specified in Table II.

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TABLE II. Mechanical properties.

Characteristics	Minimum Average	Minimum Individual
Compression strength, ultimate psi x 10^3	21	18
Compression modulus, psi x 10^6	2.0	1.8

3.4 Environmental conditions.

3.4.1 Mechanical properties at high and low temperatures. The mechanical properties shall be as specified in Table III. Requirements shall be met at the temperature specified after soaking for 30 minutes.

TABLE III. Mechanical properties at high and low temperatures.

Characteristics	Minimum Average		Minimum Individual	
	at 68.3ºC	at -53.9ºC	at 68.3ºC	at -53.9ºC
Compressive strength, ultimate psi x 10^3	12	29	9	22
Compression modulus, psi x 10^6	1.4	2.0	1.2	1.5

3.5 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in accordance with this specification. All material shall be free of dirt, foreign matter, or other contaminates (see 4.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 3.1 and 6.2)

b. Quality conformance inspection (see 4.3).

4.2.1 First article inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of not less than 10 lbs. of the material, manufactured under the same conditions as those specified for subsequent production, shall be subjected to first article inspection.

Accomplishment of first article inspection shall be as specified herein. If the first article sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained. Testing of the first article sample, to determine compliance with the characteristics listed in Table IV shall be in accordance with the corresponding test paragraphs.

TABLE IV. First article inspection.

Characteristics	Requirement paragraph	Test Paragraph
Storage life	3.2.2	4.5.4.2
Mechanical properties at high and low temperatures	3.4.1	4.5.4.3

4.3 Quality conformance inspection.

4.3.1 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be conducted on a random sample selected from each production lot in accordance with Table V.

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TABLE V. Sampling plan for quality conformance inspection.

Lot Size	Sample Size
2 to 315	All
316 to 35,000	315
35,001 to 150,000	490
150,001 and over	715
NOTE: The acceptance number in all cases is ZERO. The rejection number in all cases is ONE.	

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted at one time for acceptance inspection.

4.3.3 Quality conformance inspection. Quality conformance inspection of the sample specified in 4.3.1, to determine compliance with characteristics specified in Table VI, shall be conducted in accordance with the corresponding test and inspection paragraphs.

TABLE VI. Quality conformance inspection.

Characteristics	Requirement paragraph	Test paragraph
Composition	3.2.1	4.5.4.1
Mechanical properties	3.3.2	4.5.4.3
Resin content, % by weight	3.3.1	4.5.4.4
Volatile content, % by weight	3.3.1	4.5.4.5
Specific gravity	3.3.1	4.5.4.6
Gel time, minutes	3.3.1	4.5.4.7
Packaging, packing and marking	5.1	4.6
Workmanship	3.5	4.7

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4.4 Inspection equipment. The inspection equipment used in conducting evaluation and tests shall be as specified in the applicable test methods and procedures paragraphs (see table VI).

4.5 Test methods and procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a.	Temperature	Room ambient (16º to 32ºC)
b.	Altitude	Facility ground
c.	Humidity	Facility ambient up to 95% relative humidity

4.5.2 Test sequence. Test sequence within each classification of inspection shall be at the option of the supplier.

4.5.3 Test specimen preparation. The type and configuration of test specimens shall be as specified by the individual test paragraph. Test specimens shall be molded at 149º to 165ºC for 25 + 5 - 0 minutes at 50 ± 15 lbf/in^2 pressure to achieve a specific gravity greater than the minimum per Table I.

4.5.4 Tests.

4.5.4.1 Composition. The composition shall be certified by the supplier and the condition in the container shall be determined visually to establish conformance to 3.2.1.

4.5.4.2 Storage life. A sufficient amount of material shall be stored under the conditions specified in 3.2.2. At the end of the storage period, the material shall meet all the requirements specified herein. The supplier's certification of compliance to the storage life requirements (see 3.2.2) may be accepted during the storage period.

4.5.4.3 Compressive properties. The ultimate strength and modulus of elasticity of the material in compression shall be determined in accordance with ASTM D 695. A minimum of five specimens shall be tested at each temperature required. The material shall meet the requirements of 3.3.2 or 3.4.1, as applicable.

4.5.4.4 Resin content. Resin content of the material shall be determined in accordance with ASTM C 613. A minimum of five specimens shall be tested. The material shall meet the requirements of 3.3.1.

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4.5.4.5 Volatile content. The volatile content shall be determined as follows:

a. Specimen size shall be 10 ± 2 grams selected from random areas in the sample.

b. The specimens shall be weighed to the nearest 0.01 gram.

c. Place the specimens in crucibles in a circulating air oven preheated to 149º to 165ºC.

d. Allow the specimens to remain in the oven for 15 ± 1 minute.

e. Remove the specimens and place in a desiccator until cooled to room ambient temperature.

f. Reweigh the specimens to the nearest 0.01 gram.

g. Calculate the volatile content using the formula:

$$C_V(\%) = \frac{W_O - W_1}{W_O} \times 100$$

where:

C_V (%) = Volatile content %
W_O = Original specimen weight
W_1 = Devolatilized specimen weight

h. Report the individual and average of five specimens. The volatiles content shall be as specified in 3.3.1.

4.5.4.6 Specific gravity. Specific gravity of the material shall be determined in accordance with ASTM D 792. A minimum of five specimens shall be tested and the material shall meet the requirements of 3.3.1.

4.5.4.7 Gel time. The gel time shall be determined as follows:

a. Specimens approximately 5 grams in mass shall be selected from random areas in the sample.

b. The specimen shall be placed on a hot plate preheated to 149º to 165ºC.

c. After 3 to 5 seconds, remove the sample, leaving a small amount of resin on the surface of the hot plate.

d. Start the stop watch.

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e. Probe the resin until it will no longer form strings when the probe is removed.

f. Stop the watch and record the gel time.

g. Report the individual and average values of five tests. The material shall conform to 3.3.1.

4.6 Packaging, packing and marking. The inspector shall ascertain that packaging, packing and marking of the material shall conform to this specification.

4.7 Workmanship. The material shall be visually examined to assure conformance to workmanship requirement of 3.5.

5. PACKAGING

5.1 Preservation, packing, unitization and marking. Preservation, packing, unitization and marking shall be Level C (see 6.2).

5.1.1 Level C. Preservation, packing, unitization and marking shall be in accordance with ASTM D 3951. Additional marking shall include, but not be limited to, the following:

a. Manufacturer’s name and address

b. Manufacturer’s designation

c. Batch number

d. Date of manufacture

e. Number of this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material is intended for use as compression molded, high strength plastic parts.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. When a first article is required (see 3.1 and 4.2)

d. Container size and quantity required (see 4.5.4.1)

e. Requirement for certification of composition (see 4.5.4.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Composites
Compression forming
Plastic

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity
Army - MI

Review activity
DLA - GS

Project Number 9330-A043

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